

# Appendix A - South-west Corner Marine Park (western arm) Natural Values Report

South-west Marine Parks Network

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## Natural Values

Understanding of natural values

## Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone.

| Date         | Campaign name                          | Method | Areas                         | Samples | Habitat | Fish  | Length |
|--------------|----------------------------------------|--------|-------------------------------|---------|---------|-------|--------|
|              |                                        |        |                               |         | Count   | Count |        |
| Jun 2020     | 2020-06 south-west stereo-BRUVs        | BRUV   | NPZ                           | 42      | 40      | 39    | 33     |
| Oct-Nov 2020 | 2020-10 south-west stereo-BRUVs        | BRUV   | Coastal waters, NPZ, SPZ      | 275     | 273     | 272   | 244    |
| Oct-Nov 2020 | 2020-10 south-west BOSS                | BOSS   | Coastal waters, NPZ, SPZ      | 262     | 262     | -     | -      |
| Mar 2021     | 2021-03 West-Coast BOSS                | BOSS   | Coastal waters, NPZ           | 154     | 154     | -     | -      |
| Mar 2023     | 2023-03 Augusta stereo-BOSS            | BOSS   | Coastal waters, NPZ, SPZ      | 73      | 70      | -     | -      |
| Mar 2023     | 2023-03 SwC BOSS                       | BOSS   | NPZ                           | 88      | 88      | -     | -      |
| Mar 2023     | 2023-03 SwC stereo-BRUVs               | BRUV   | NPZ                           | 32      | 32      | 32    | 32     |
| Oct 2024     | 2024-10 SWC stereo-BRUVs               | BRUV   | Coastal waters, NPZ, SPZ      | 190     | 190     | 188   | 178    |
| Nov 2024     | 2024-11 SwC stereo-BOSS                | BOSS   | Coastal waters, MUZ, NPZ, SPZ | 137     | 117     | -     | -      |
| Mar-Apr 2025 | 2025-04 South-west Corner BOSS         | BOSS   | NPZ, SPZ                      | 210     | 210     | -     | -      |
| Apr 2025     | 2025-04 South-west-corner stereo-BRUVs | BRUV   | NPZ, SPZ                      | 150     | 148     | 143   | 137    |

## Benthic ecosystem extent

### Observations

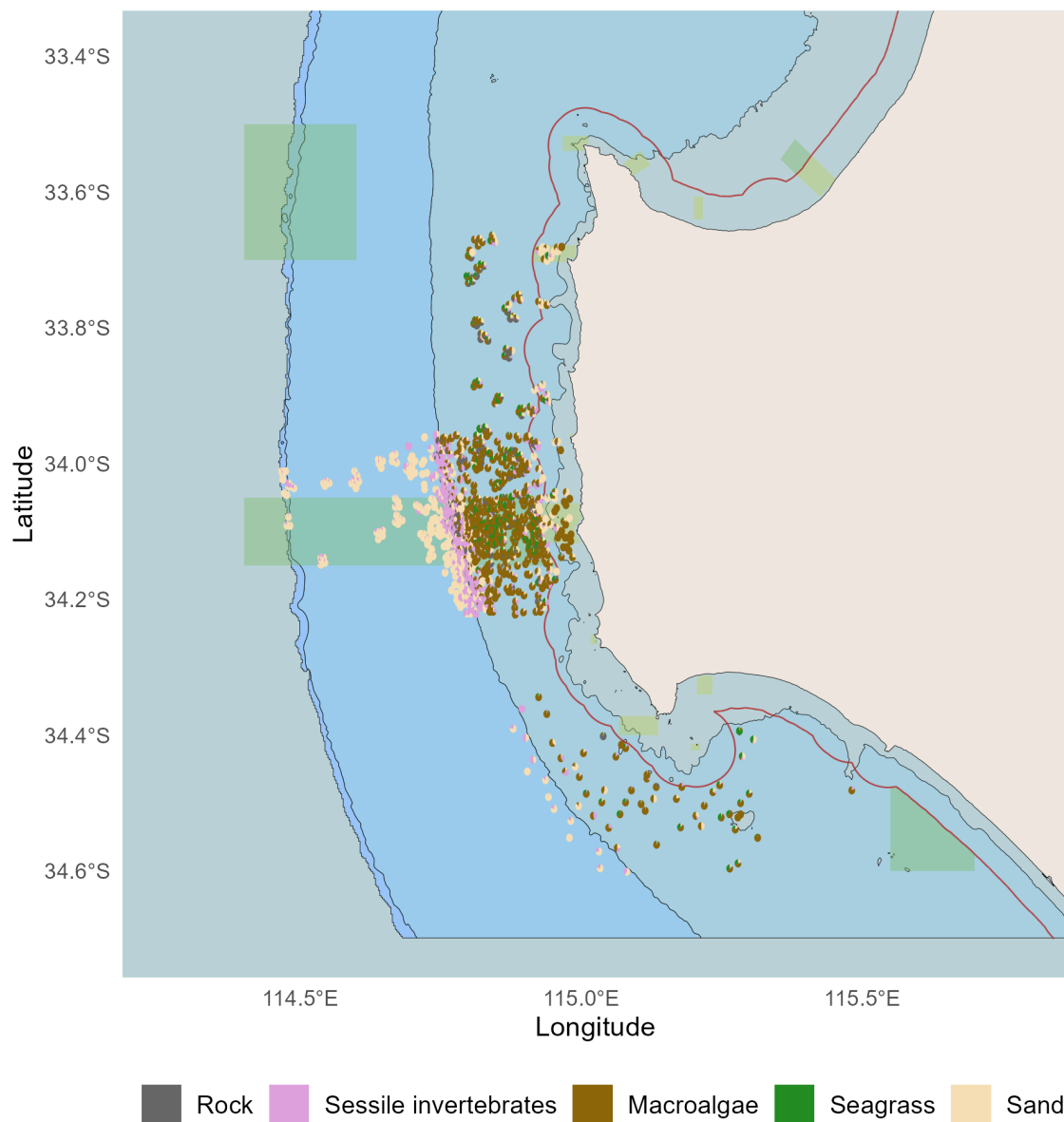


Figure 1: Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Dominant habitat types

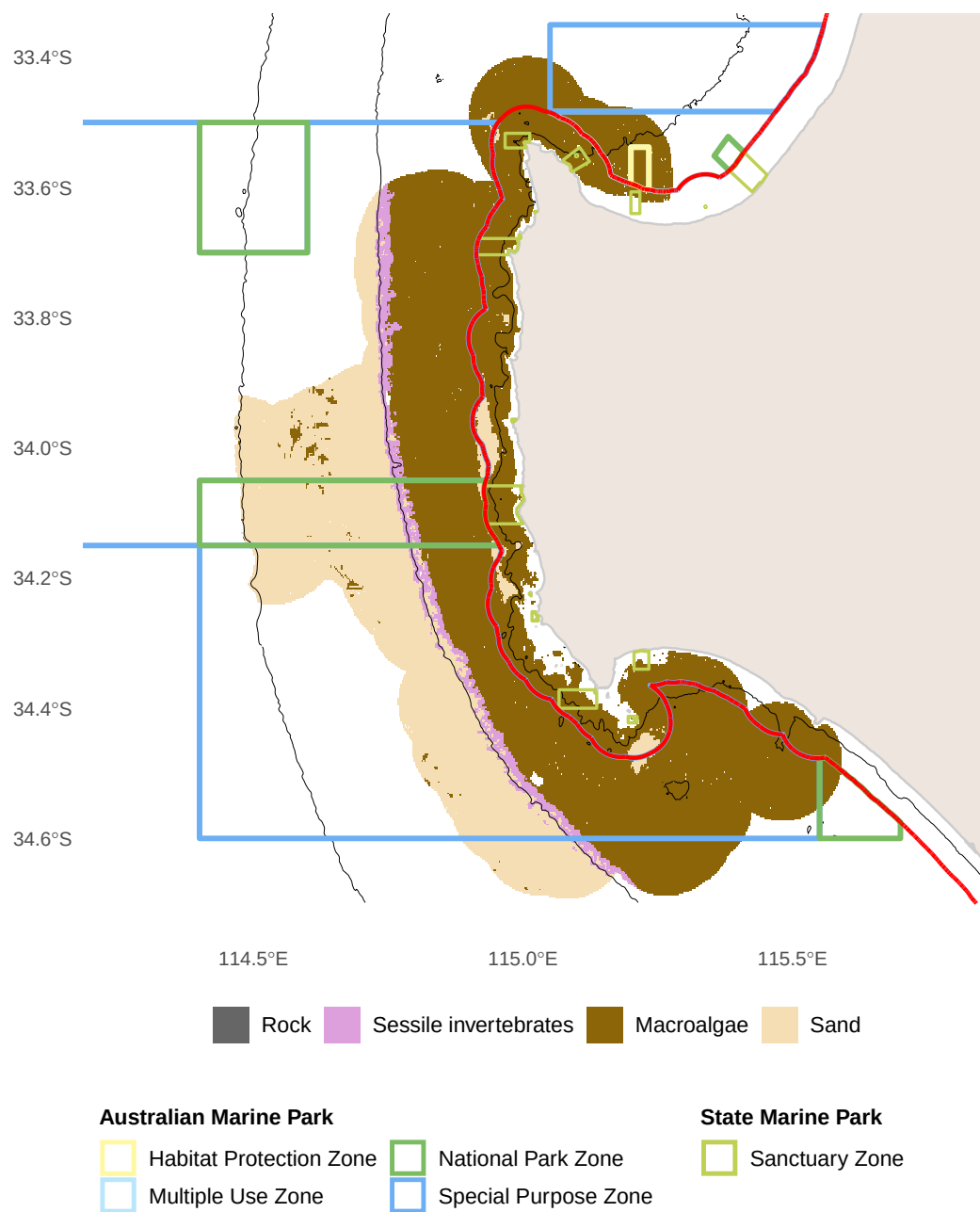


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Habitat distribution and probability

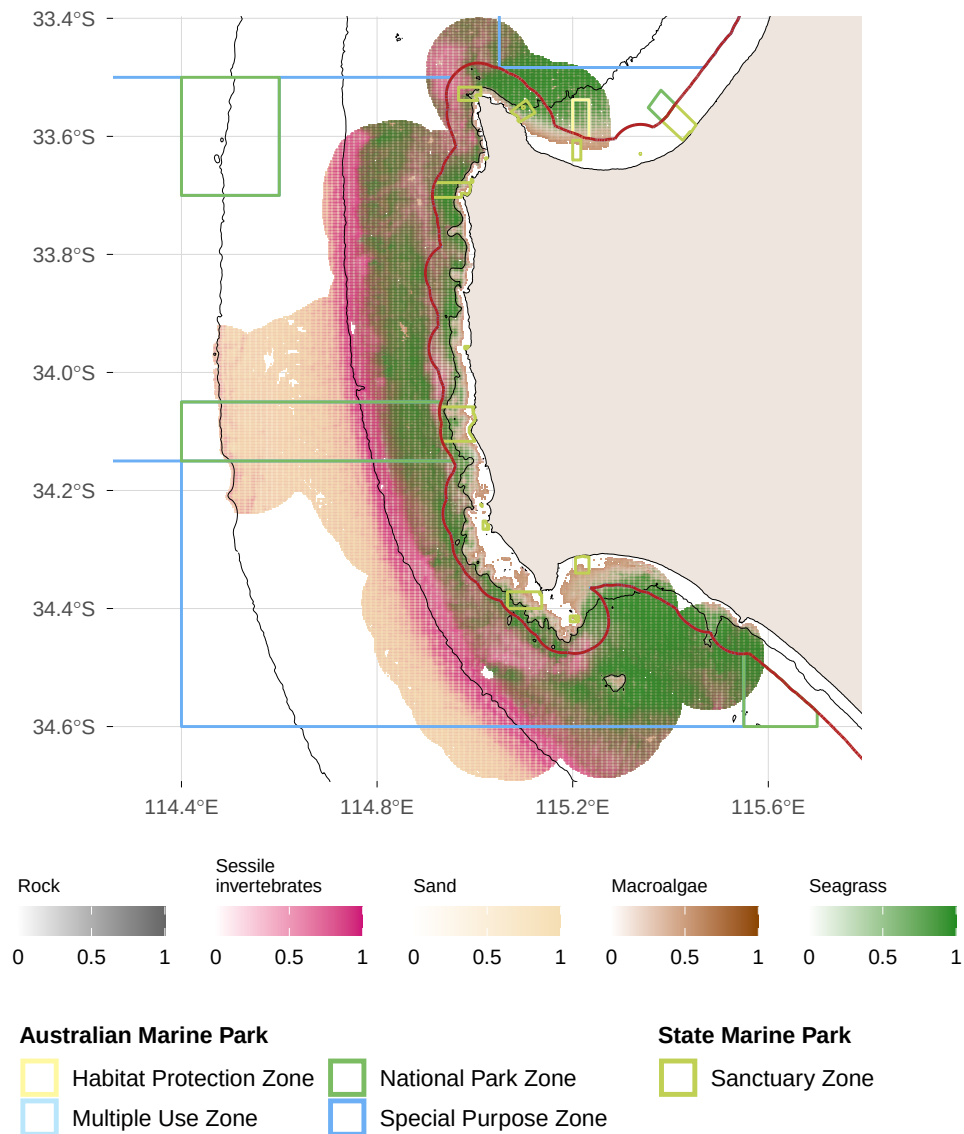


Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Functional reef distribution and probability

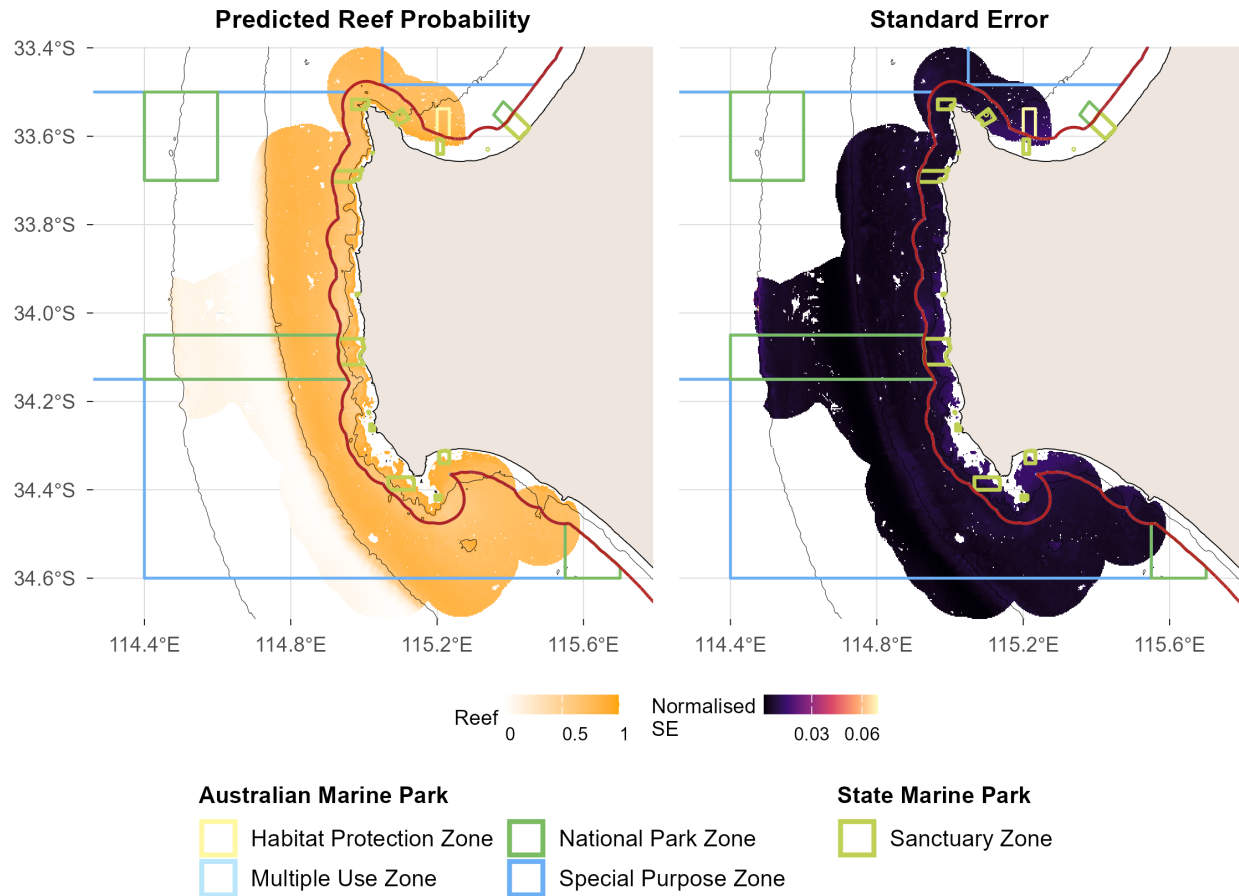


Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Benchmarks of benthic natural values

### Sand

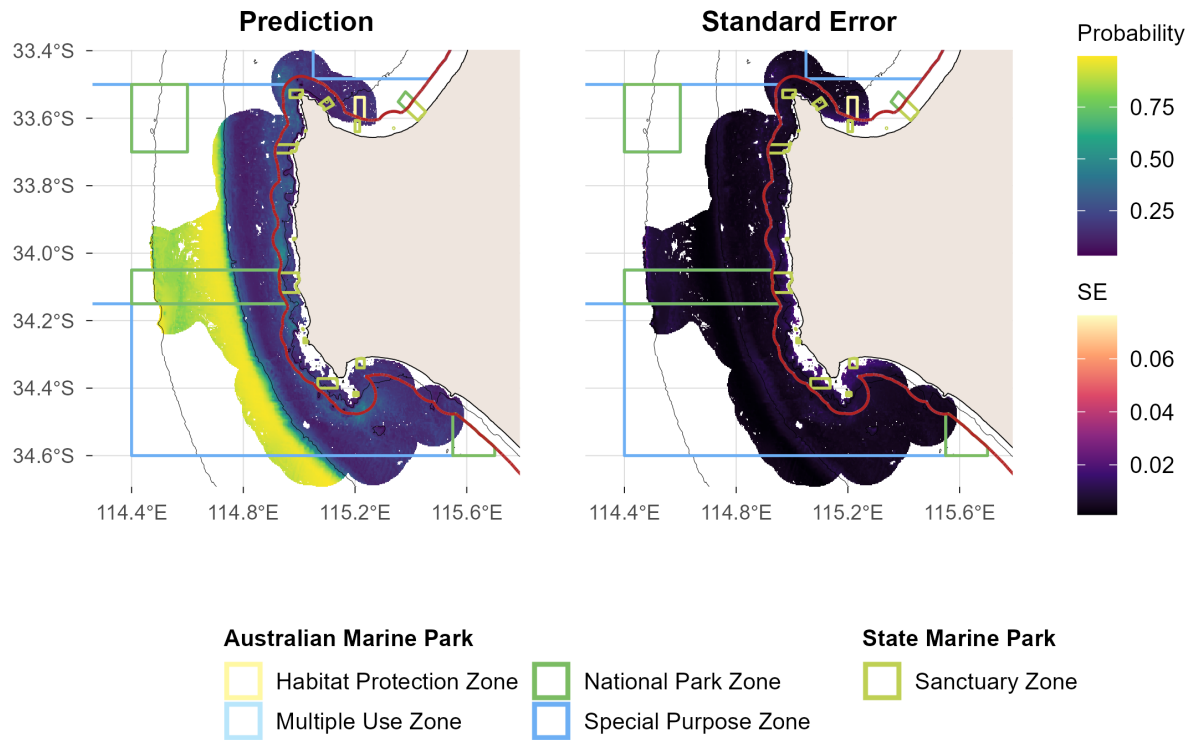


Figure 5.1: Sand: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Macroalgae

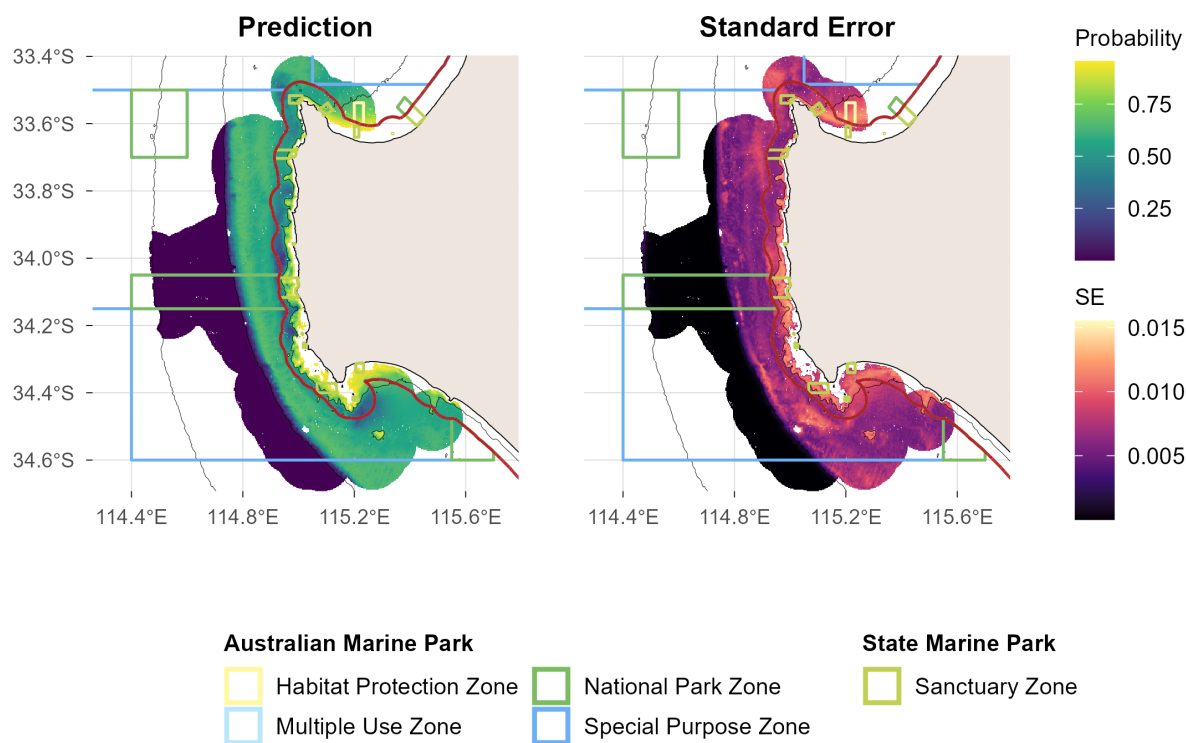


Figure 6.1: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Seagrass

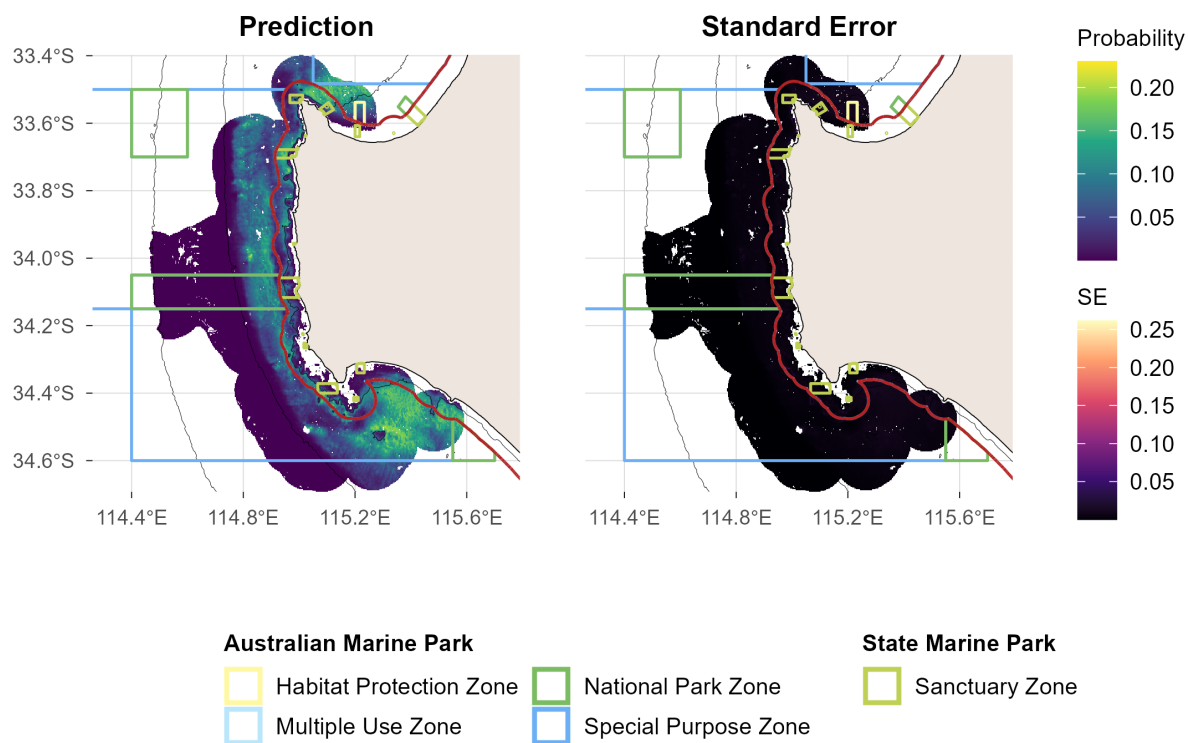


Figure 7.1: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Sessile invertebrates

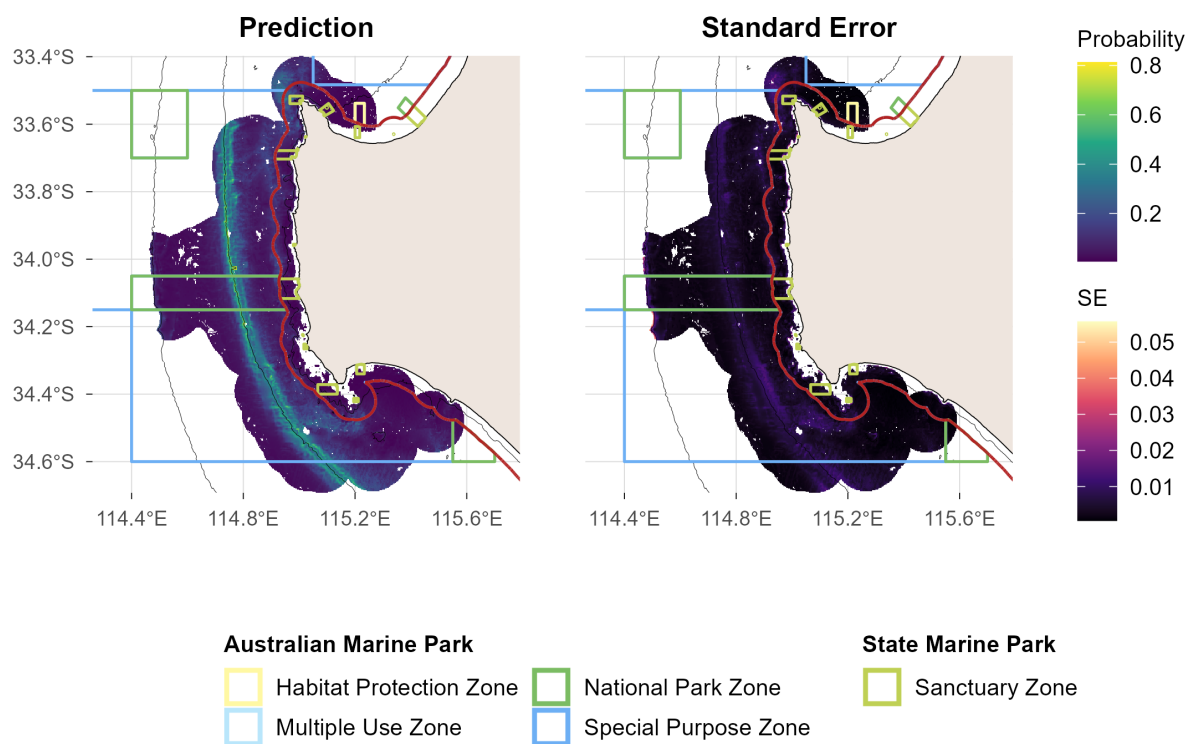


Figure 8.1: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Rock

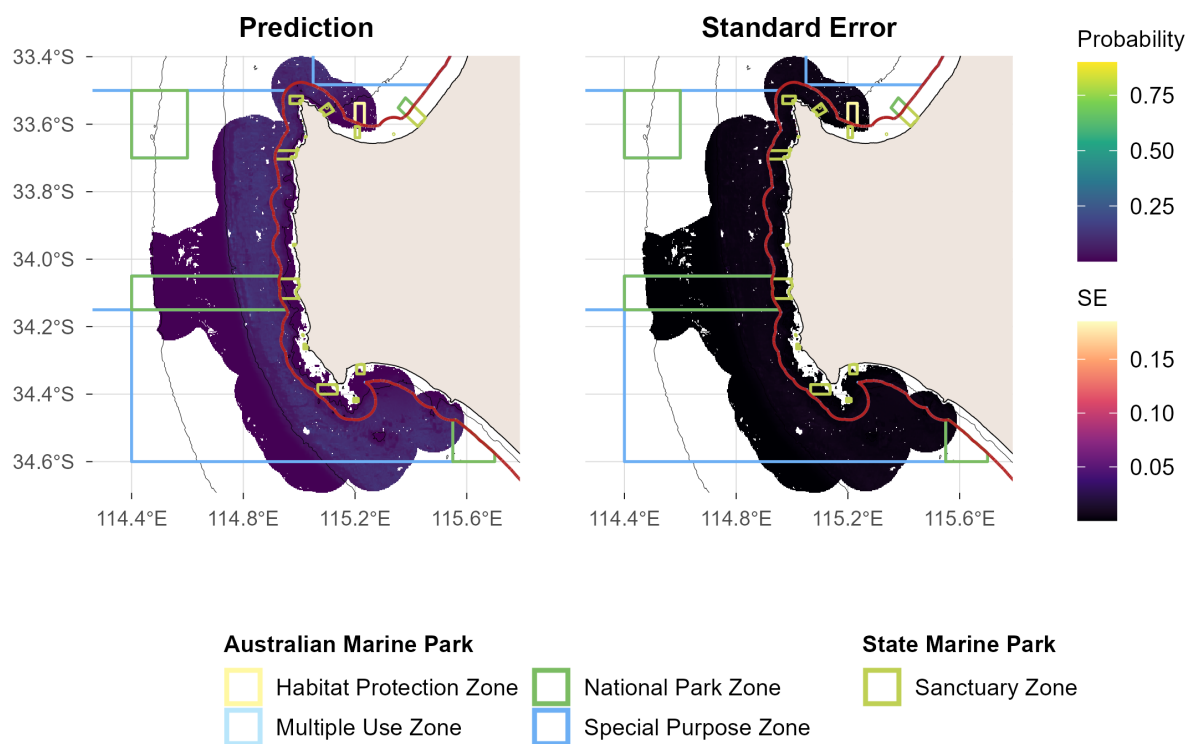
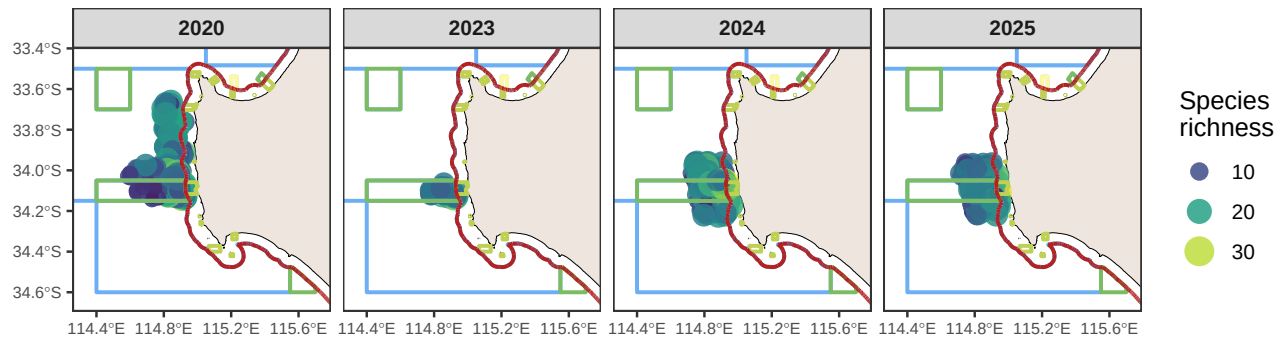


Figure 9.1: Rock: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of demersal fish natural values

Survey effort

a)



b)

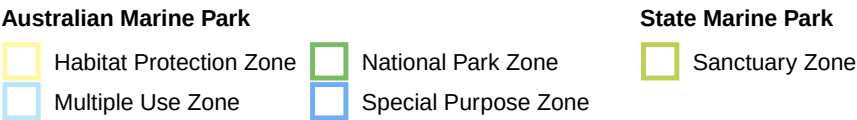
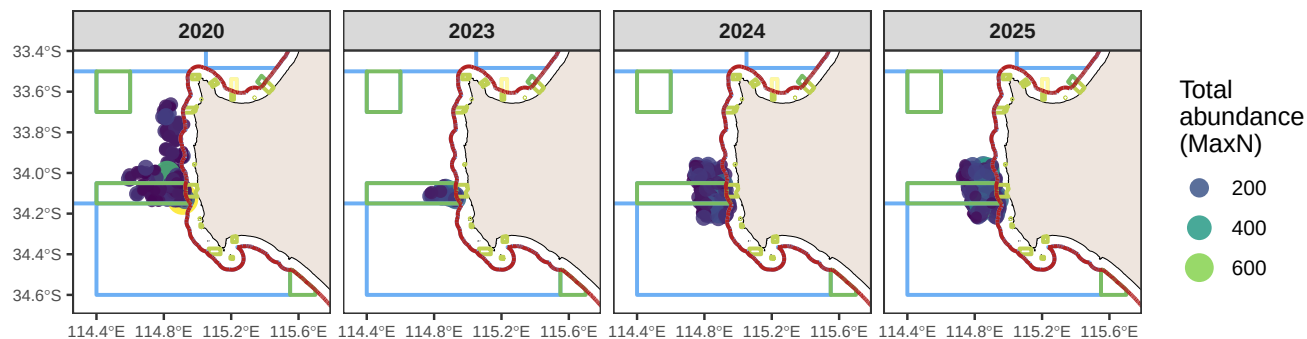


Figure 10: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

## Species richness

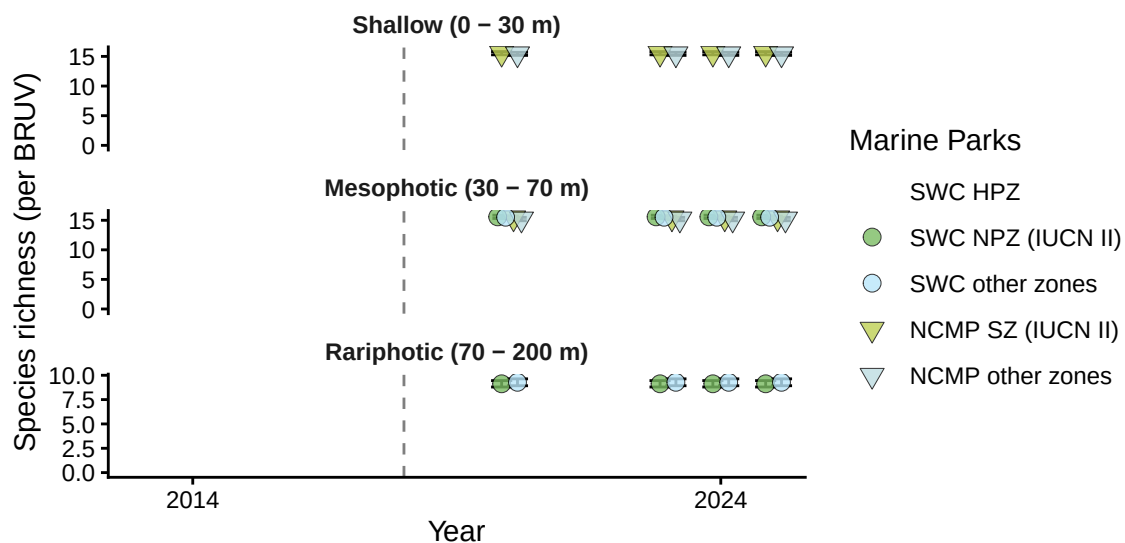


Figure 11.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

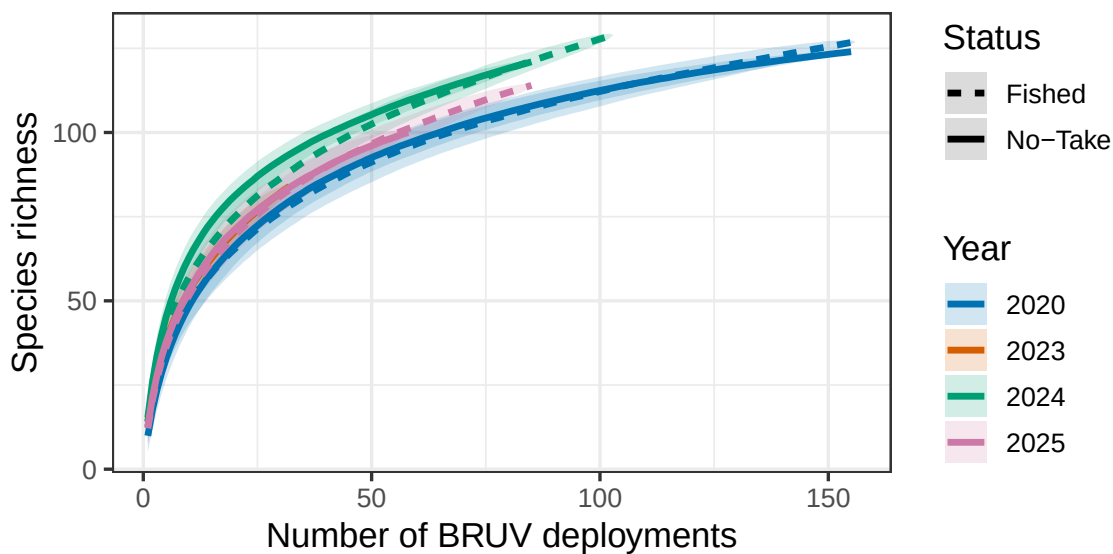


Figure 11.2: Demersal fish - Species richness: species accumulation curve by year from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Species richness

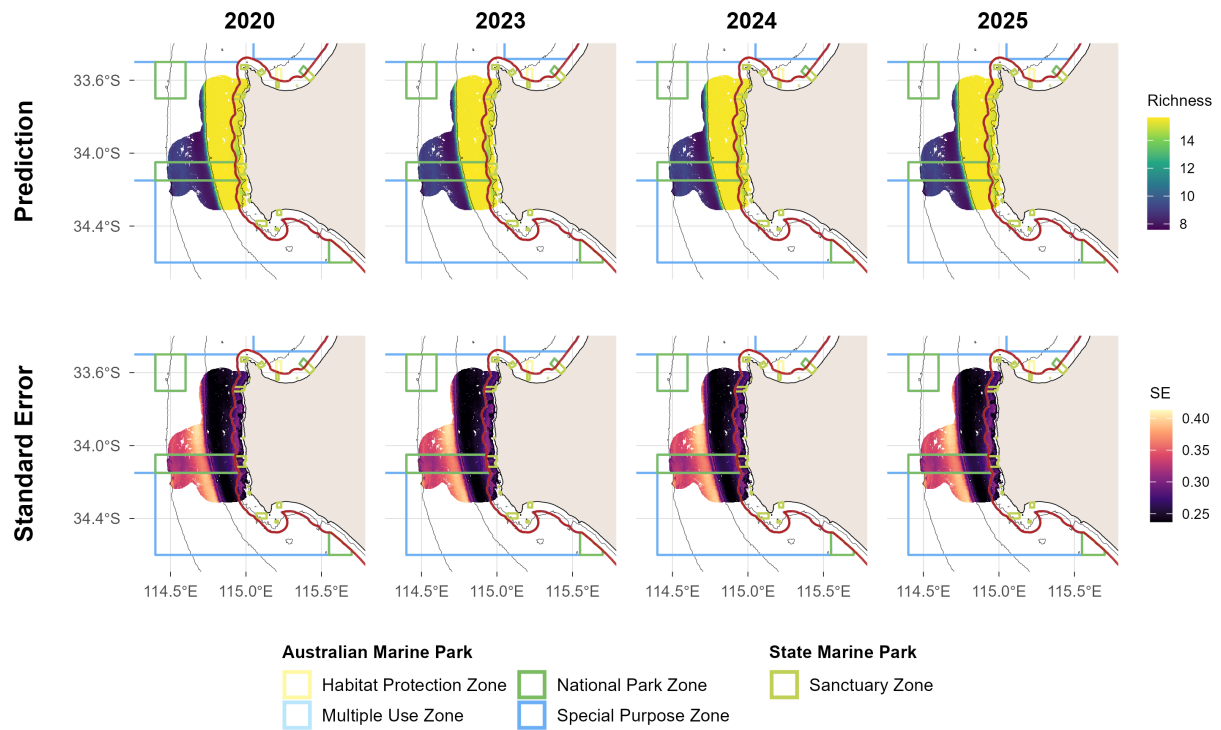


Figure 11.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

## Total abundance

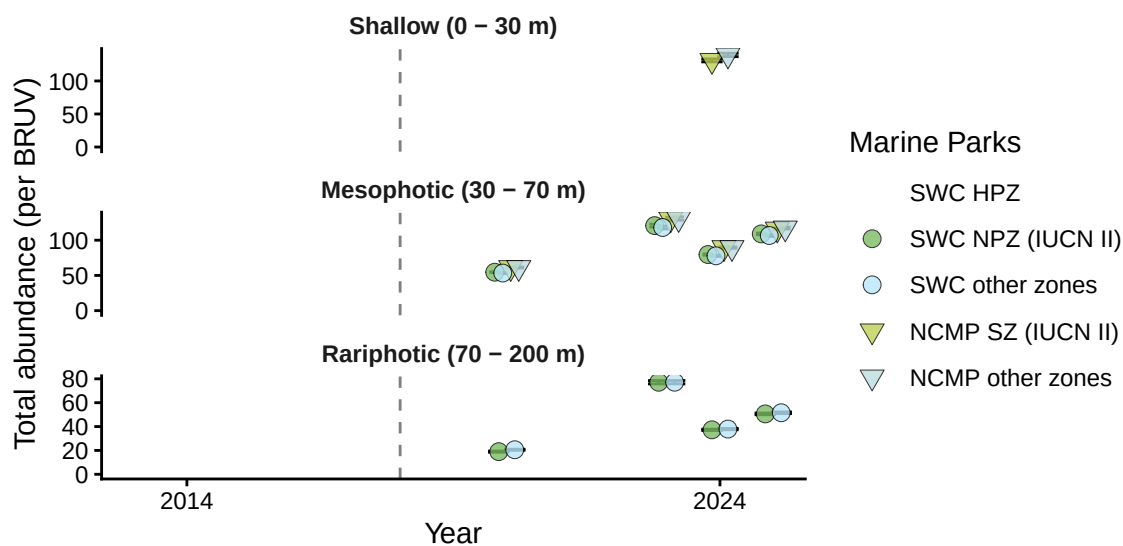


Figure 12.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

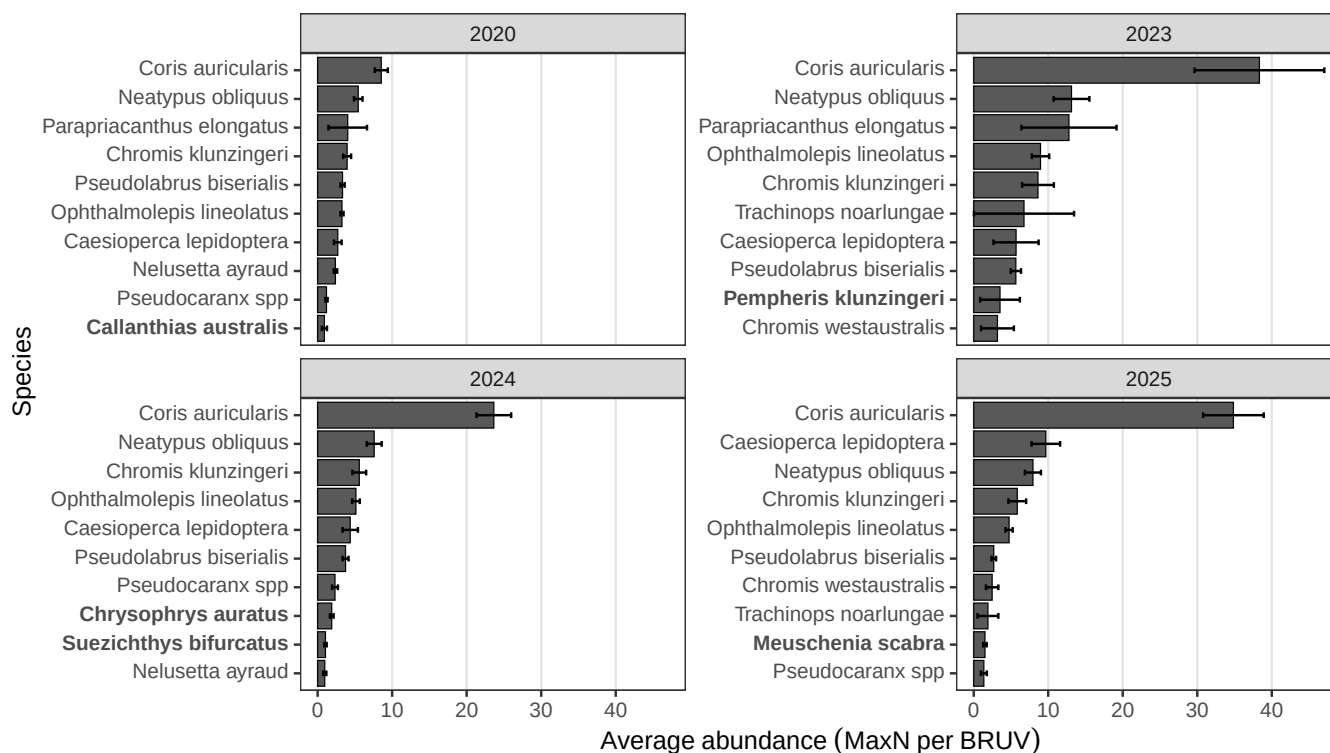


Figure 12.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Total abundance

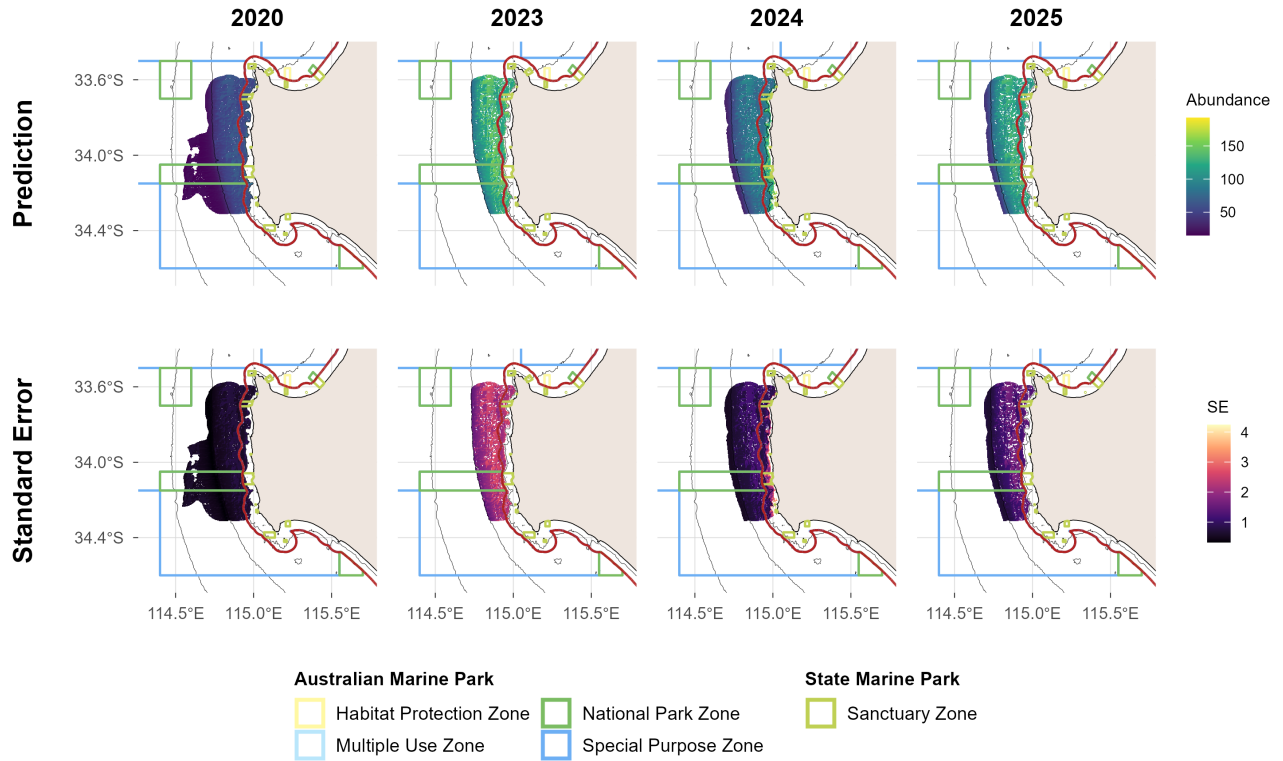


Figure 12.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index\*

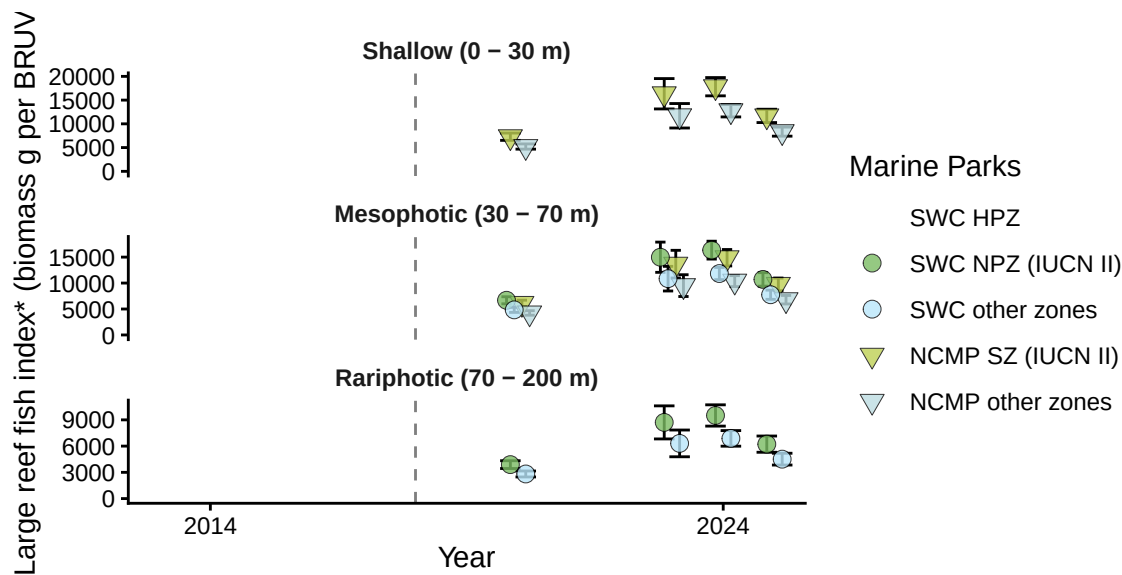


Figure 13.1: Demersal fish - Large Reef Fish Index\*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth). \*Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

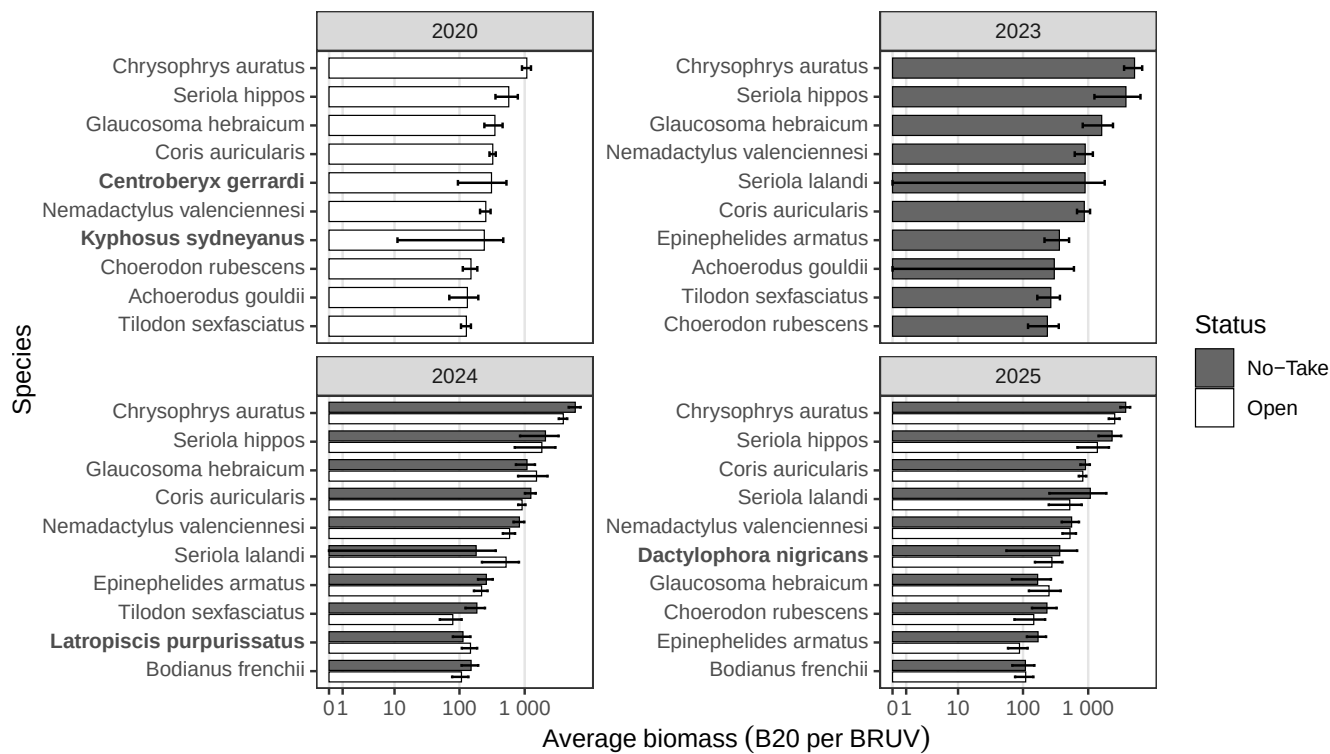


Figure 13.2: Demersal fish - Large Reef Fish Index\*: top 10 greatest biomass species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Large Reef Fish Index\*

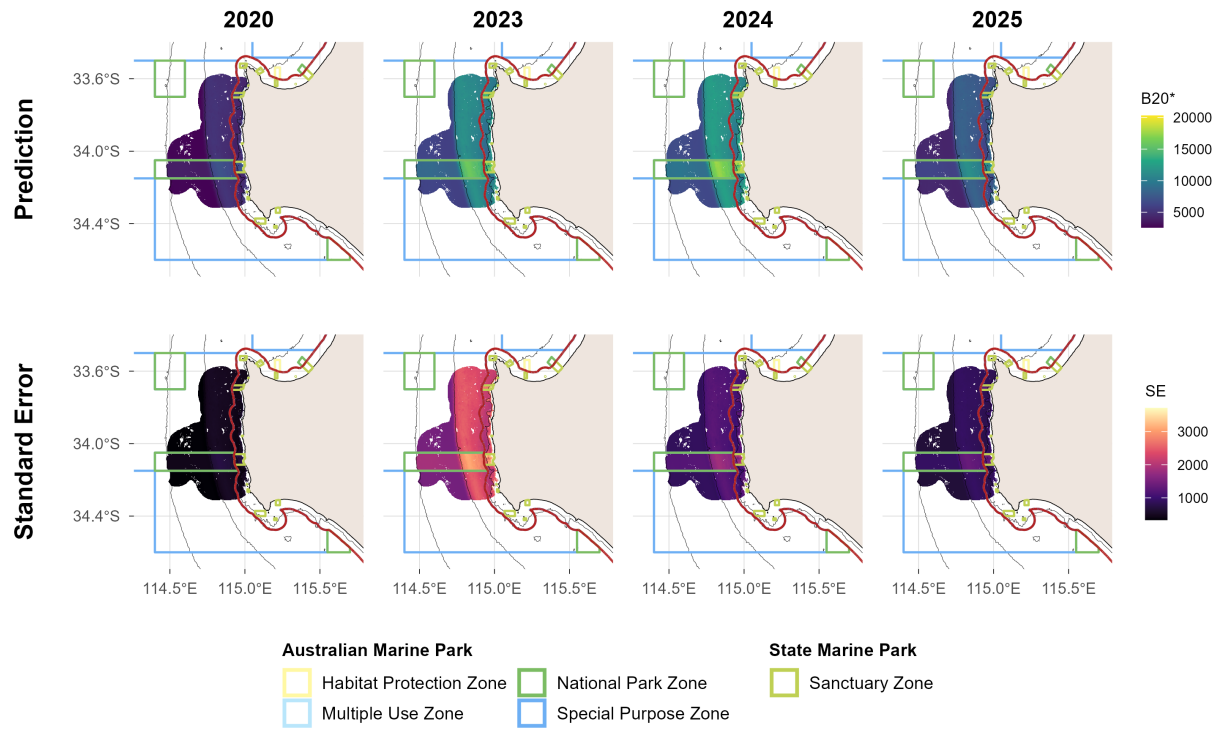


Figure 13.3: Demersal fish - Large Reef Fish Index\*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

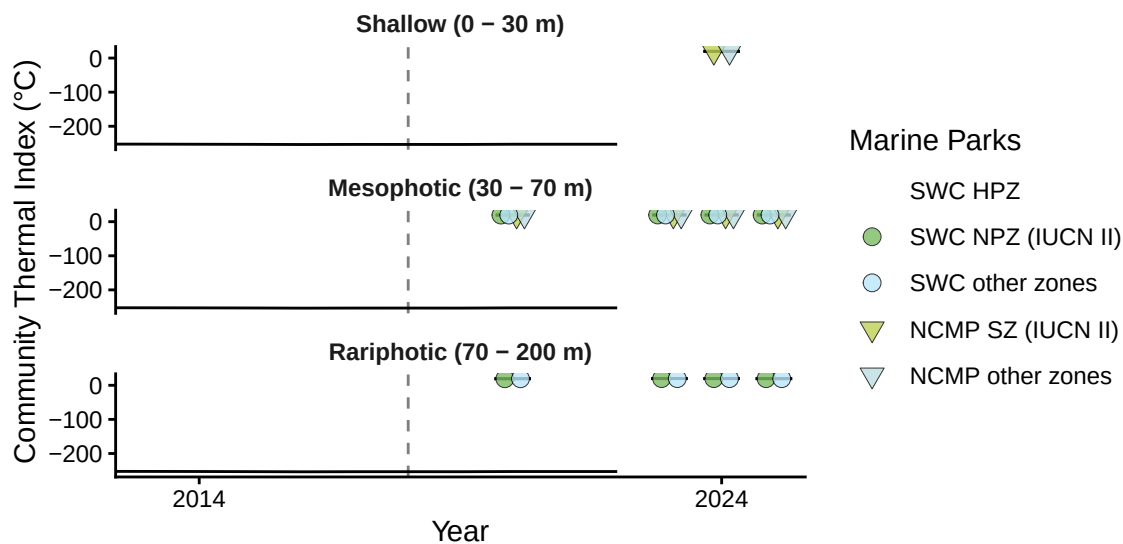


Figure 14.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

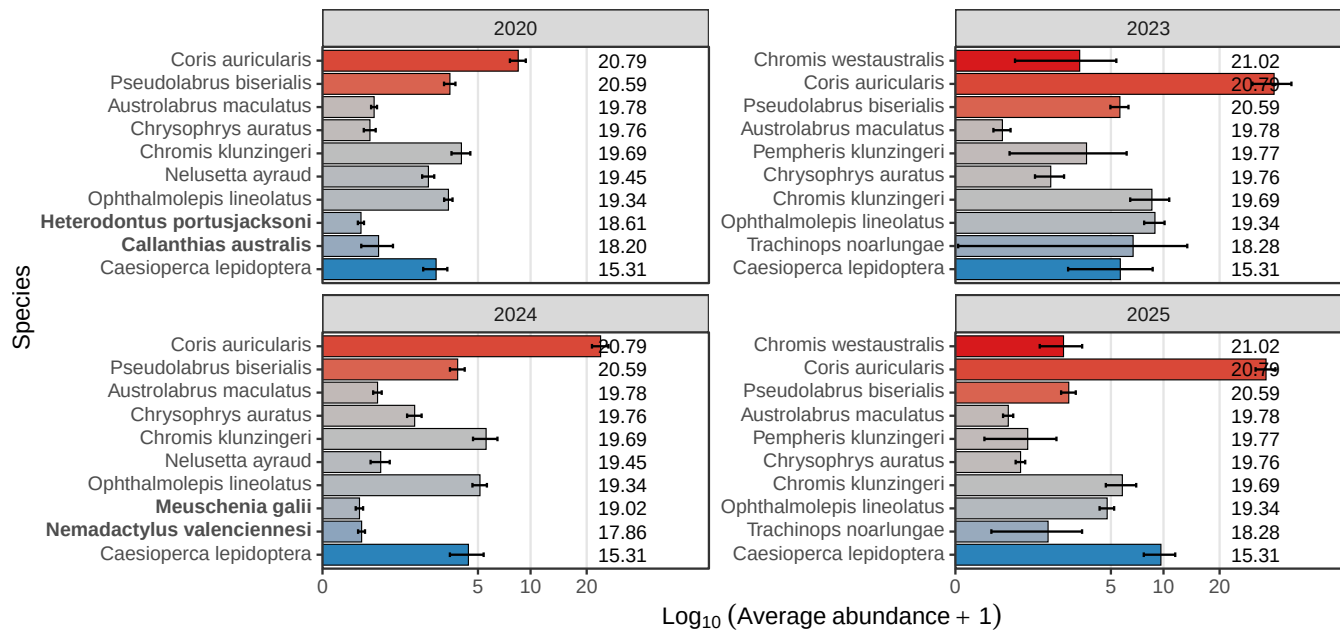


Figure 14.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. SWC = South-west Corner Marine Park (Commonwealth).

## Reef fish thermal index

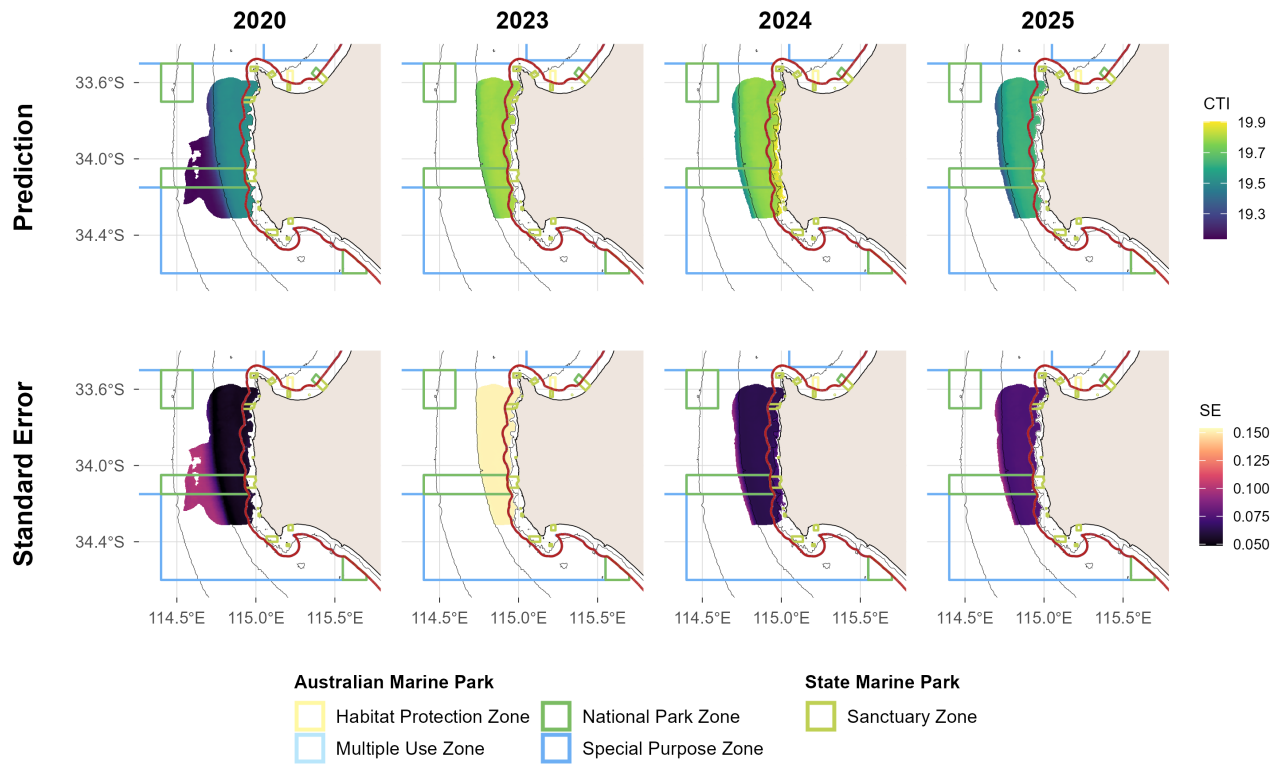


Figure 14.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.